College Club Management System (CCMS)

IA3 Software Engineering Project — Detailed Software Test Cases & Verification Report

Framework: Python pytest 9.1.1 **Test Cases:** 30 Executed **Pass Rate:** 100.0% (30 Pass, 0 Fail) **Date:** November 2026

1. Positive Testing Suite (Valid Inputs & Successful Operations)

Test ID	Req ID	Objective / Description	Input / Test Data	Expected Result	Actual Outcome	Status
TC-POS-01	FR-CLUB-01	Verify landing page loads successfully with HTTP 200.	GET /	HTTP 200 OK; React DOM root loaded.	HTTP 200 OK in 12ms.	**PASS**
TC-POS-02	FR-CLUB-02	Verify public clubs directory route /clubs is accessible.	GET /clubs	HTTP 200 OK; Content-Type: text/html.	HTTP 200 OK; HTML served.	**PASS**
TC-POS-03	FR-AUTH-02	Verify user login route /login is accessible.	GET /login	HTTP 200 OK; sign-in form rendered.	HTTP 200 OK; sign-in rendered.	**PASS**
TC-POS-04	FR-AUTH-01	Verify student registration route /register is accessible.	GET /register	HTTP 200 OK; registration rendered.	HTTP 200 OK; form rendered.	**PASS**
TC-POS-05	FR-CLUB-02	Verify Supabase fetch of active categories.	GET /rest/v1/club_categories	HTTP 200 OK; JSON array with categories.	HTTP 200 OK; Academic, Cultural.	**PASS**
TC-POS-06	FR-CLUB-01	Verify Supabase fetch of active clubs.	GET /rest/v1/clubs	HTTP 200 OK; JSON array with active clubs.	HTTP 200 OK; CSS, Music Club.	**PASS**
TC-POS-07	NFR-PERF-01	Verify Vite bundles and serves static CSS/JS chunks.	GET / (parse script tags)	HTML references valid compiled script asset.	Script reference verified.	**PASS**

2. Negative Testing Suite (Invalid Inputs & Unexpected Conditions)

Test ID	Req ID	Objective / Description	Input / Test Data	Expected Result	Actual Outcome	Status
TC-NEG-01	FR-AUTH-02	Verify sign-in fails with invalid password.	POST /auth/token (bad pw)	HTTP 400 Bad Request; error returned.	HTTP 400; 'Invalid login credentials'.	**PASS**
TC-NEG-02	FR-AUTH-02	Verify sign-in rejects nonexistent email accounts.	POST /auth/token (ghost user)	HTTP 400 Bad Request; login rejected.	HTTP 400 Bad Request returned.	**PASS**
TC-NEG-03	FR-AUTH-04	Verify unauthenticated client cannot alter user roles.	PATCH /profiles role='admin'	RLS blocks update; 0 rows modified or 401.	HTTP 200 []; 0 rows updated.	**PASS**
TC-NEG-04	FR-ATT-05	Verify unauthenticated client cannot insert attendance.	POST /rest/v1/attendance	HTTP 401/403 Forbidden; insert rejected.	HTTP 401/403; row insert blocked.	**PASS**
TC-NEG-05	FR-CLUB-04	Verify nonexistent club slug returns empty array.	GET /clubs?slug=bogus-slug	HTTP 200 OK with empty array [].	HTTP 200 OK; [] returned.	**PASS**

3. Boundary Testing Suite (Minimum, Maximum & Edge Cases)

Test ID	Req ID	Objective / Description	Input / Test Data	Expected Result	Actual Outcome	Status
TC-BND-01	FR-REG-03	Verify capacity check with zero capacity (capacity=0).	count=0, capacity=0	Registration rejected (False).	Evaluated to False; rejected.	**PASS**
TC-BND-02	FR-REG-03	Verify boundary when count reaches exact capacity.	count=99,100,101 vs cap=100	99 -> True; 100,101 -> False.	Exact limit strictly enforced.	**PASS**
TC-BND-03	FR-CLUB-03	Verify search handler with empty string boundary.	GET /clubs?name=ilike.%%	HTTP 200 OK; returns all clubs.	HTTP 200 OK; all clubs returned.	**PASS**
TC-BND-04	FR-CLUB-03	Verify search with extreme string length (1000 chars).	GET /clubs?name=ilike.<1000 A's>	Handled without crash/buffer overflow.	HTTP 200 OK; [] in 48ms.	**PASS**
TC-BND-05	NFR-SEC-06	Verify malformed non-UUID input boundary.	GET /clubs?id=eq.invalid-uuid	HTTP 400/422; PostgreSQL rejects format.	HTTP 400 Bad Request (invalid UUID).	**PASS**

4. Functional Testing Suite (Core Features & Business Logic)

Test ID	Req ID	Objective / Description	Input / Test Data	Expected Result	Actual Outcome	Status
TC-FNC-01	FR-CLUB-02	Verify club category referential integrity.	Query clubs & categories	All club category_id link to active categories.	100% referential integrity verified.	**PASS**
TC-FNC-02	FR-ATT-04	Verify mathematical attendance percentage formula.	compute_rate(present/total)	Perfect: 100%, Mixed: 75%, Empty: 0%.	100.0%, 75.0%, 0.0% verified.	**PASS**
TC-FNC-03	FR-MEM-01	Verify membership lifecycle states domain.	Domain set validation	{pending, approved, rejected} allowed.	Valid states confirmed.	**PASS**
TC-FNC-04	FR-ATT-02	Verify attendance status domain values.	Domain set validation	{present, absent, late} allowed.	Exact domain verified.	**PASS**
TC-FNC-05	FR-AUTH-03	Verify role permissions matrix (Student/Coord/Admin).	RBAC policy matrix	Strict isolation across user classes.	Zero privilege leakage.	**PASS**

5. Integration Testing Suite (Module Interoperability)

Test ID	Req ID	Objective / Description	Input / Test Data	Expected Result	Actual Outcome	Status
TC-INT-01	NFR-PERF-02	Verify bidirectional client-to-backend PostgREST API.	Ping Frontend & Supabase	Both services return HTTP 200 OK.	Frontend 200, PostgREST 200.	**PASS**
TC-INT-02	5.4	Verify PostgREST returns required CORS headers.	OPTIONS /rest/v1/clubs	HTTP 200/204; Access-Control-Allow-Origin.	HTTP 204 with valid CORS.	**PASS**
TC-INT-03	6.4	Verify foreign key constraint rejects orphaned category.	POST club with bogus cat_id	HTTP 400/409 FK violation or 401 RLS block.	Rejected; orphan insert blocked.	**PASS**
TC-INT-04	FR-AUTH-02	Verify Supabase GoTrue Auth service health.	GET /auth/v1/health	HTTP 200 OK; GoTrue status confirmed.	HTTP 200 OK; service verified.	**PASS**

6. Validation & Error Handling Suite

Test ID	Req ID	Objective / Description	Input / Test Data	Expected Result	Actual Outcome	Status
TC-VAL-01	FR-AUTH-01	Verify signup rejects short passwords (<6 chars).	POST /auth/signup pw='123'	HTTP 400/422; password length error.	HTTP 422: 'At least 6 characters'.	**PASS**
TC-VAL-02	FR-AUTH-02	Verify empty JSON auth payload returns 400.	POST /auth/token with {}	HTTP 400 Bad Request.	HTTP 400 Bad Request returned.	**PASS**
TC-VAL-03	NFR-REL-02	Verify malformed JSON syntax is caught gracefully.	POST malformed JSON string	HTTP 400 Bad Request; server does not crash.	HTTP 400; graceful parser error.	**PASS**
TC-VAL-04	NFR-USE-04	Verify arbitrary unknown routes return SPA fallback.	GET /missing/route/xyz	HTTP 200 OK; HTML root for client routing.	HTTP 200 OK; clean SPA fallback.	**PASS**

Total Executed: 30 **Passed:** 30 **Failed:** 0 **Evaluation Decision: APPROVED FOR IA3 DEMO**